NEURAL NETWORKS AND DEEP LEARNING

CHALLENGE BASED LEARNING (CBL)

Image Classification Challenge

BIRD SPECIES RECOGNITION

BY:

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BIRD SPECIES RECOGNITION

1. INTRODUCTION

1.1 Background

Birds play an important role in maintaining ecological balance and biodiversity. Identifying bird species is useful for wildlife monitoring, ecological research, and conservation activities. However, manually identifying different species can be difficult because many birds have similar colours, shapes, and feather patterns.

Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have made automated image classification more effective. This project uses a pretrained ResNet-18 model to classify images into 200 bird species using the CUB-200-2011 dataset. It also investigates whether cropping the bird region improves classification performance.

1.2 Problem Statement

Bird species recognition is challenging because different species may have similar visual characteristics, and image backgrounds can contain irrelevant information. The project aims to develop a deep learning-based classification system for 200 bird species and compare the performance of original images with bounding-box-cropped images.

1.3 Motivation

The project is motivated by the need for automated bird identification to support biodiversity studies and wildlife observation. It also investigates whether focusing on the bird region instead of the complete image can improve classification performance.

1.4 Objectives

The main objectives of the project are:

1. To develop a deep learning model for recognizing 200 bird species.
2. To use the CUB-200-2011 dataset for training and evaluation.
3. To apply transfer learning using a pretrained ResNet-18 model.
4. To compare classification using original images and cropped bird regions.
5. To evaluate both approaches using accuracy, macro precision, macro recall, and macro F1-score.

1.5 Scope of the Project

The project focuses on image-based classification of 200 bird species using Python, PyTorch, and Google Colab. It includes image preprocessing, model training, bounding-box cropping, and performance evaluation.

The system is evaluated using the official test set of the CUB-200-2011 dataset. Real-time bird detection and field deployment are outside the scope of the current implementation.

2. PROPOSED METHODOLOGY

2.1 System Overview

The proposed system classifies bird images into 200 species using a pretrained ResNet-18 deep learning model. Two experiments are conducted: one using original images and another using bird regions cropped with bounding-box annotations.

The workflow includes dataset preparation, image preprocessing, model training, testing, and performance comparison.

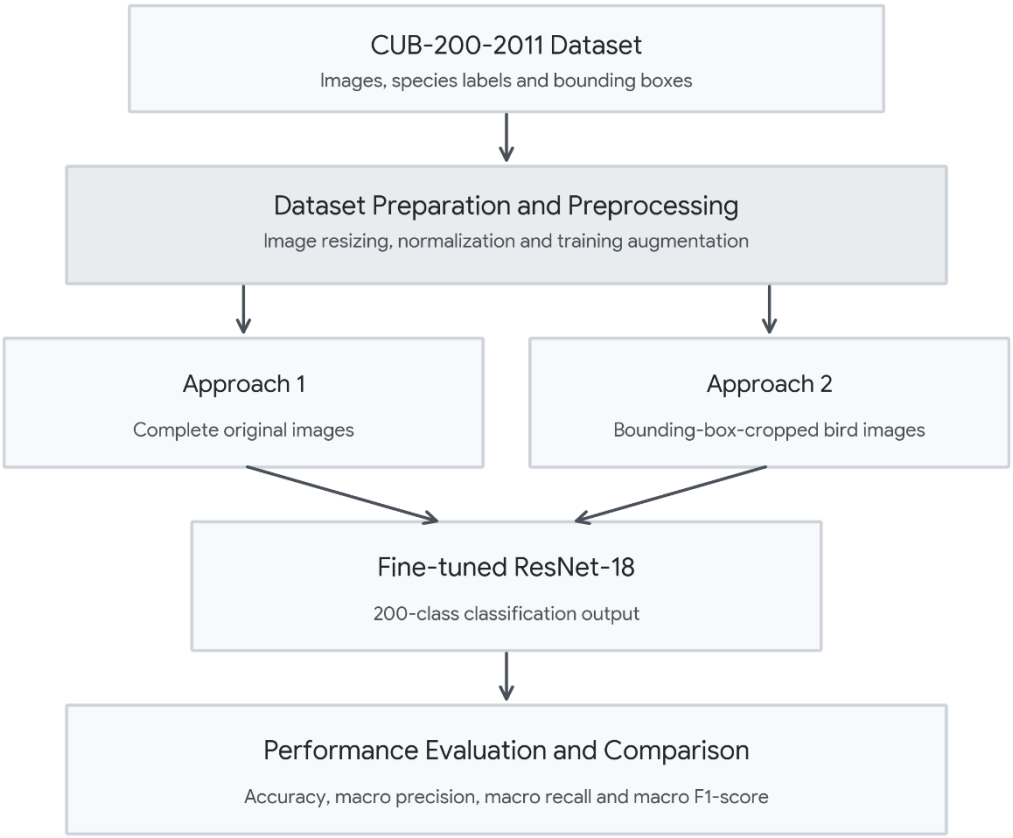


Figure 1: *Workflow of the proposed bird species recognition system*

2.2 Dataset Description

The CUB-200-2011 dataset is used for this project. It contains 11,788 images belonging to 200 bird species, along with class labels and bounding-box annotations.

Table 1: Dataset distribution

Dataset subset	Number of images
Official training set	5,994
Training subset	Approximately 5,094
Validation subset	Approximately 900
Official test set	5,794
Total	11,788

The official training set is divided into training and validation subsets using an 85:15 split. The official test set is reserved for final evaluation.

2.3 Image Preprocessing

All images are converted to RGB format and resized to (224 \times 224) pixels. Pixel values are normalized using ImageNet channel statistics. Random horizontal flipping is applied during training.

For the cropped-image experiment, bounding-box coordinates are used to extract the bird region before resizing and normalization.

2.4 Model Architecture and Training

A pretrained ResNet-18 model is used for feature extraction and classification. Its final fully connected layer is replaced with a 200-class output layer.

Table 2: Training configuration

Parameter	Value
Model	ResNet-18
Pretrained weights	ImageNet

Output classes	200
Batch size	32
Optimizer	Adam
Loss function	Cross-entropy
Initial learning rate	0.0001
Weight decay	0.0001
Training duration	5 epochs per model

The original-image model used a reduced learning rate of 0.00005 for its final three epochs. The cropped-image model used a learning rate of 0.0001 for all five epochs. The best validation checkpoint was retained for each model.

2.5 Evaluation Metrics

The models are evaluated using accuracy, macro precision, macro recall, and macro F1-score.

- **Accuracy:** Proportion of correctly classified images.
- **Precision:** Measures how often predictions for a class are correct.
- **Recall:** Measures how many actual examples of a class are identified correctly.
- **F1-score:** Harmonic mean of precision and recall.

Macro-averaged precision, recall, and F1-score give equal importance to each of the 200 species.

3. IMPLEMENTATION AND RESULTS

3.1 Implementation Environment

The project was implemented in Google Colab using Python, PyTorch, Torchvision, Pandas, Scikit-learn, Matplotlib, and Seaborn.

The dataset was loaded using its metadata files, and two ResNet-18 models were trained: one with original images and another with bounding-box-cropped images.

3.2 Experimental Results

Both models were evaluated on the official test set containing 5,794 images.

Table 3: Test performance comparison

Metric	Original images	Cropped images
Accuracy	65.98%	**76.49%**
Macro precision	67.70%	**77.91%**
Macro recall	66.27%	**76.60%**
Macro F1-score	65.77%	**76.21%**

The cropped-image model achieved a test accuracy improvement of 10.51 percentage points over the original-image model.

3.3 Accuracy Comparison

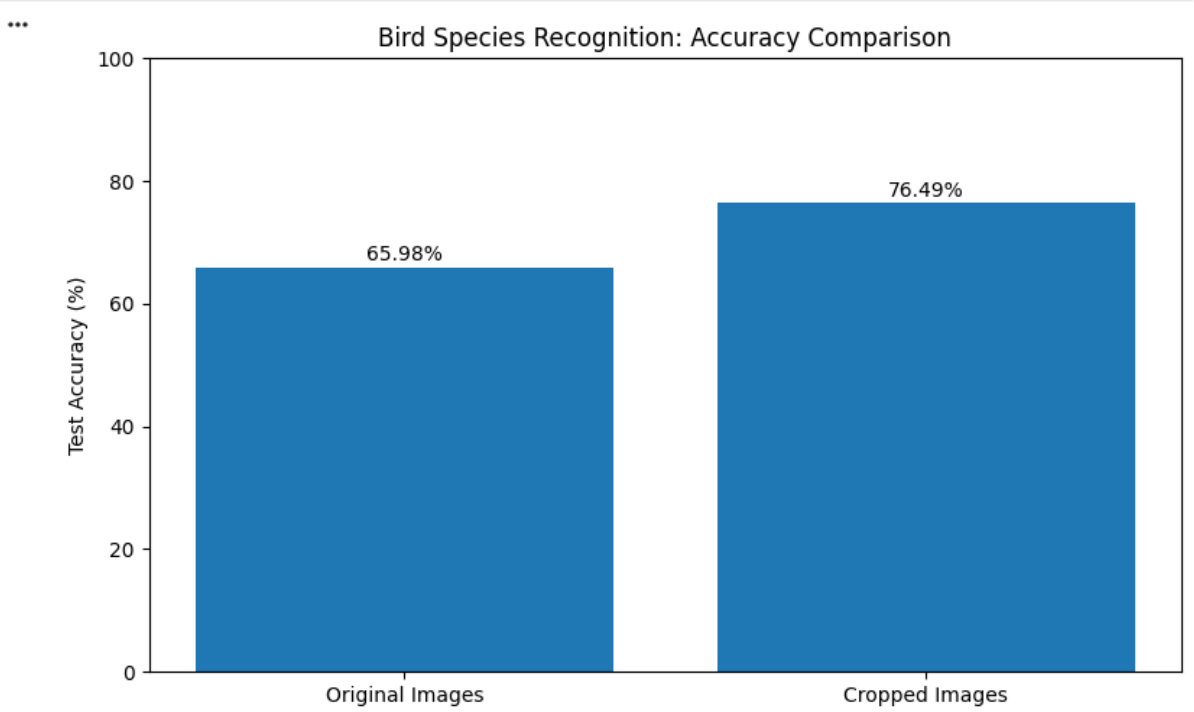


Figure 2: *Test accuracy comparison between original and cropped images*

The graph shows that the cropped-image model achieved 76.49% accuracy, compared with 65.98% for original images. This suggests that focusing on the bird region can improve classification performance.

3.4 Performance Metrics Comparison

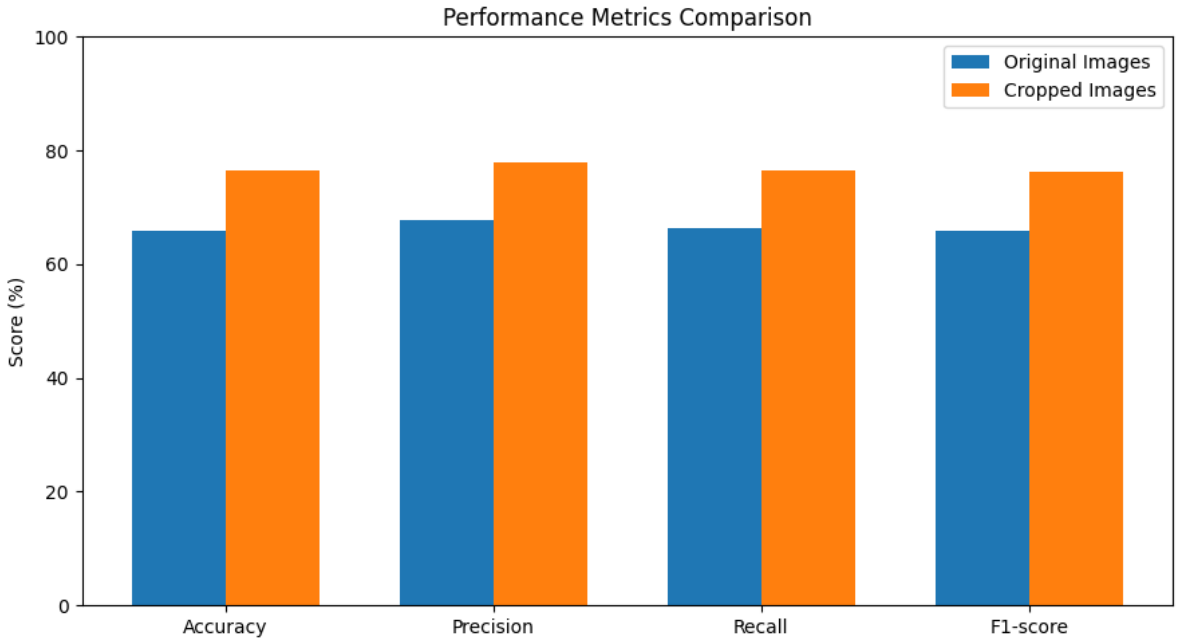


Figure 3: *Comparison of classification performance metrics*

The cropped-image model achieved higher precision, recall, and F1-score in addition to higher accuracy. These results indicate improved overall classification performance in this experiment.

3.5 Sample Prediction Analysis

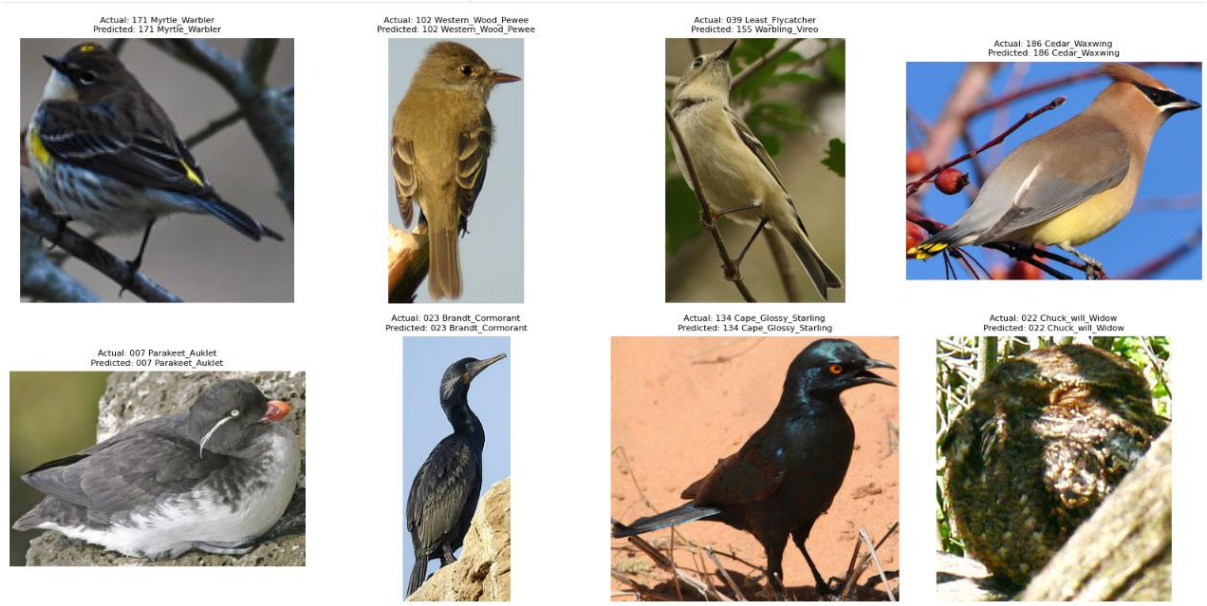


Figure 4: *Sample bird species predictions*

The sample predictions show both correct and incorrect classifications. Some visually similar species were confused, illustrating the challenges of fine-grained

bird recognition. The displayed samples are illustrative and do not replace the full test evaluation.

3.6 Confusion Matrix Analysis

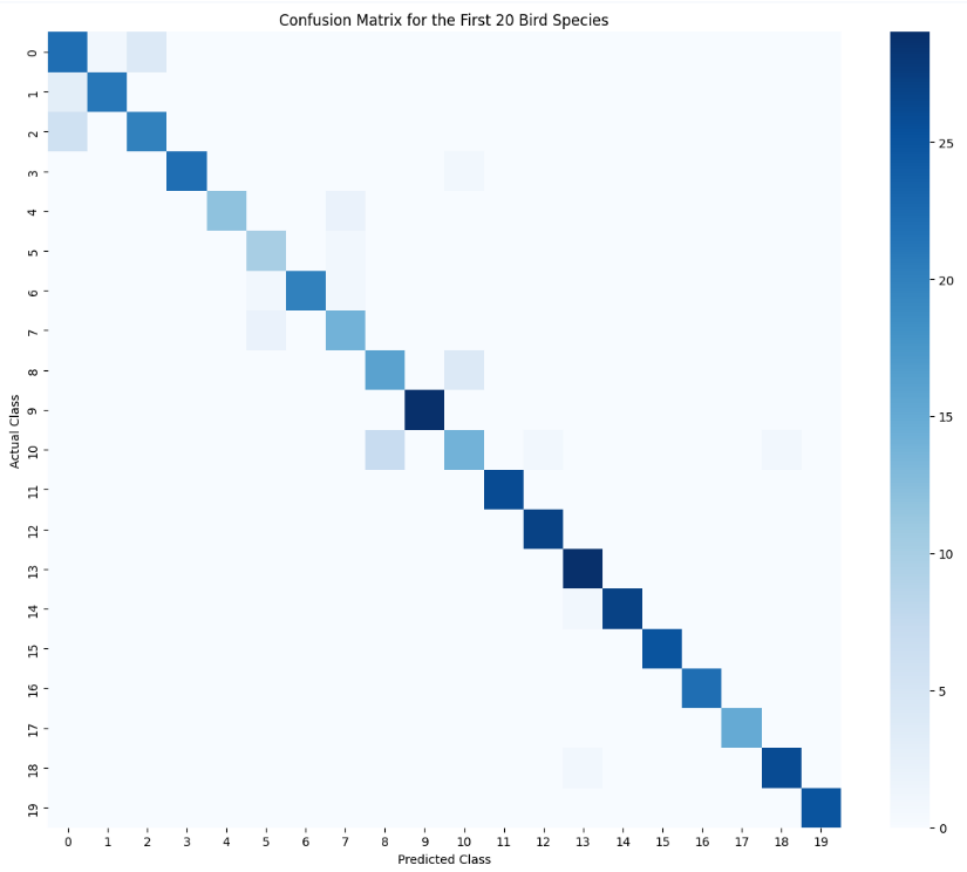


Figure 5: *Confusion matrix for the first 20 bird species*

The confusion matrix compares actual and predicted labels for the first 20 species. Values on the main diagonal indicate correct predictions, while off-diagonal values indicate misclassifications. The full classification report evaluates all 200 species.

3.7 Discussion

The cropped-image model performed better across all four evaluation metrics. Bounding-box cropping may reduce irrelevant background information and help the model focus on useful features such as feather patterns, body shape, and colour.

However, the models used different learning-rate schedules. Therefore, the improvement cannot be attributed to cropping alone with complete certainty. Further controlled experiments would help validate this finding.

4. CONCLUSION AND FUTURE WORK

4.1 Conclusion

This project developed a bird species recognition system using the CUB-200-2011 dataset and a pretrained ResNet-18 model. Two approaches were evaluated: classification using original images and classification using bounding-box-cropped bird regions.

The original-image model achieved 65.98% test accuracy, while the cropped-image model achieved 76.49%, an improvement of 10.51 percentage points. The cropped-image model also achieved higher macro precision, recall, and F1-score.

The results suggest that focusing on the bird region can improve classification performance for the implemented system. The project demonstrates the practical use of transfer learning, image preprocessing, and performance evaluation in fine-grained image classification.

4.2 Limitations

- Models were trained for only five epochs.
- Different learning-rate schedules were used.
- The cropped-image approach depends on bounding-box annotations.
- Visually similar species can still be misclassified.
- Performance on images outside the dataset has not been evaluated.

4.3 Future Work

Future improvements may include longer training, controlled comparisons, advanced architectures such as ResNet-50 or EfficientNet, improved data augmentation, automated bird detection, and deployment through a web or mobile application.